
The Inflationary Impacts of the Data-center Build-out

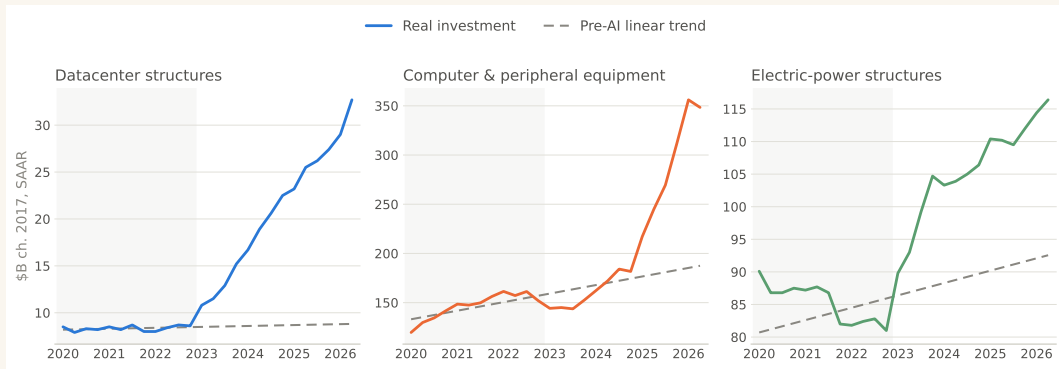
Emma Leonard, Robbie Minton, Jacob Orchard

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September 18, 2026

Board Inflation Workshop

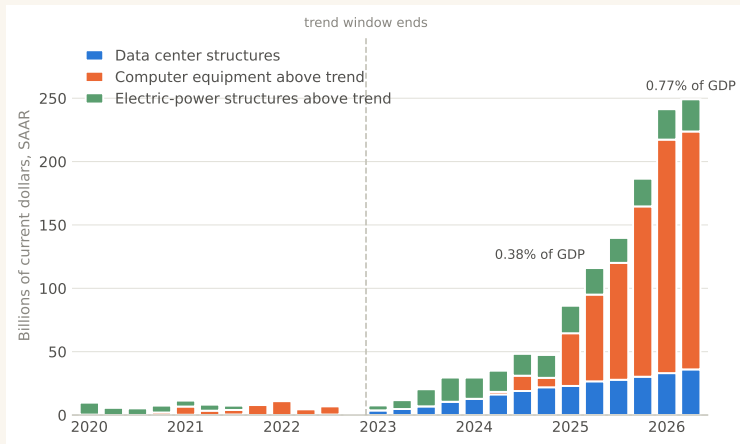
Data-center Growth has Exploded



Source: BEA and Author's Calculations. Trend estimated through 2022Q4.

- Follows (Brandsaas, Molina, Kurtzman, Nichols, Zyttek 2026)

Imputed Data Center Investment



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2. Estimate “relative” increases in price for each of these sectors assuming constrained supply
 - **Supply is less-elastic:** Base supply elasticity on (Boehm and Pandalai-Nayar 2022) estimates of highly constrained supply

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 - **Downstream Cost Effect:** Standard network downstream cost propagation

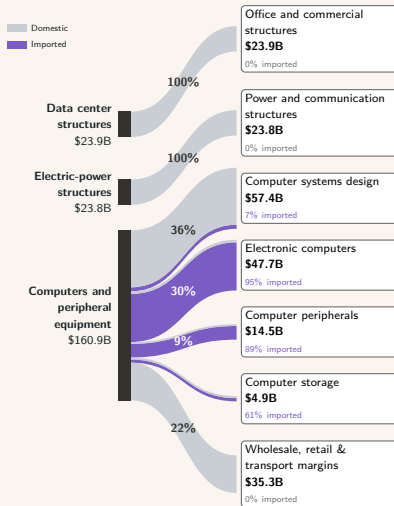
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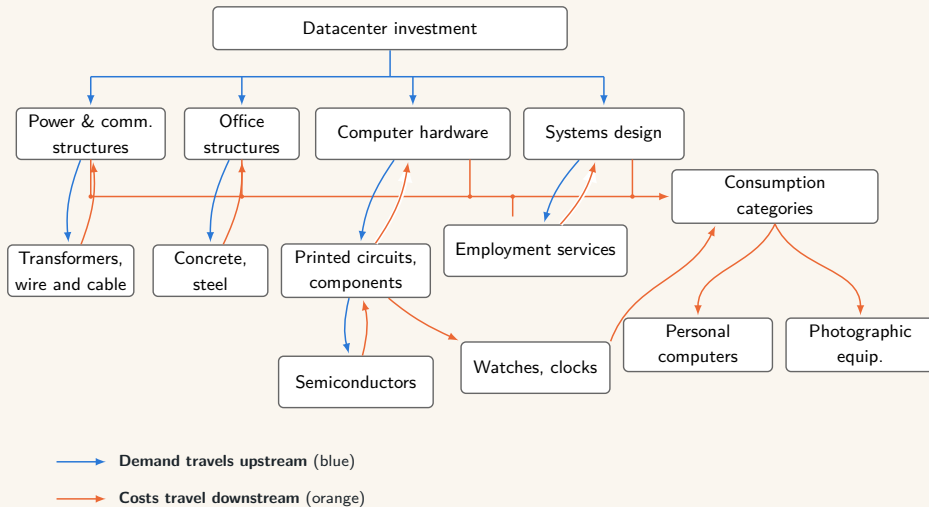
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3. Empirically test demand and price exposure in the data
 - Large increase in relative employment and prices for exposed sectors since January 2025.

Where does demand go?



Model Preview



Modeling Data Center Exposure

Total demand X_i for industry i is sum of intermediate, final demand, and new data-center demand (E):

$$X_i = \sum_j X_{ji} + D_i + E_i$$

Production Function: For all sectors $i \in \{1, \dots, N\}$:

$$Y_i = A_i F_i(X_{i1}, \dots, X_{iN}, VA_i)$$

- Sectoral production is *Leontief* in all inputs.
- VA_i : Sector i 's value-added inputs.

Log-linearizing

Log-linearizing, market clearing and using our leontieff assumption we get that:

$$y_i = \sum_j \frac{P_i X_{ji}}{P_i Y_i} y_j + \frac{P_i D_i}{P_i Y_i} d_i + \varepsilon_i$$

where

$$\varepsilon_i \equiv \frac{P_i \Delta E_i}{P_i Y_i}$$

- To account for global markets, Y_i is set as the sector's total domestic use

Moving to a Global Market

- We assume that the global production function is the same as the domestic
- Data-center demand shock is happening in other countries, but to a lesser degree
 - US has $\sim 45\%$ of world datacenters (δ), but only absorbs $\sim 34\%$ world high-tech output ($\tilde{s}_i$) (IEA, IDC).
- The US data-center shock relative to high-tech absorption is larger than the global average. We scale the high-tech shock down to make it more reflective of the global data-center demand shock:

$$\varepsilon_i^w = \frac{\tilde{s}_i}{\delta} \varepsilon_i$$

- The shock for traded tech commodities is scaled down by 24%. Nontraded investment (structures) remain unscaled ($\tilde{s}_i = \delta = 1$).

Demand Exposure

Stacking across sectors $i \in \{1, \dots, N\}$, we have

$$\mathbf{y} = \Phi \mathbf{y} + \Phi_D \mathbf{d} + \boldsymbol{\varepsilon},$$

where Φ is a matrix with entries $\frac{P_i X_{ji}}{P_i Y_i}$ where the numerator of each entry is the transpose of the BEA Input-Output use matrix.

We assume that the data-center demand shock does not directly affect final demand:

$$\mathbf{y} = (\mathbf{I} - \Phi)^{-1}(\boldsymbol{\varepsilon}).$$

Prices

Network cost pass-through (α_i = value-added share):

$$\mathbf{p} = (\mathbf{I} - \mathbf{\Omega})^{-1} \text{diag}(\alpha_i) \mathbf{w}$$

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Value added imperfectly supplied

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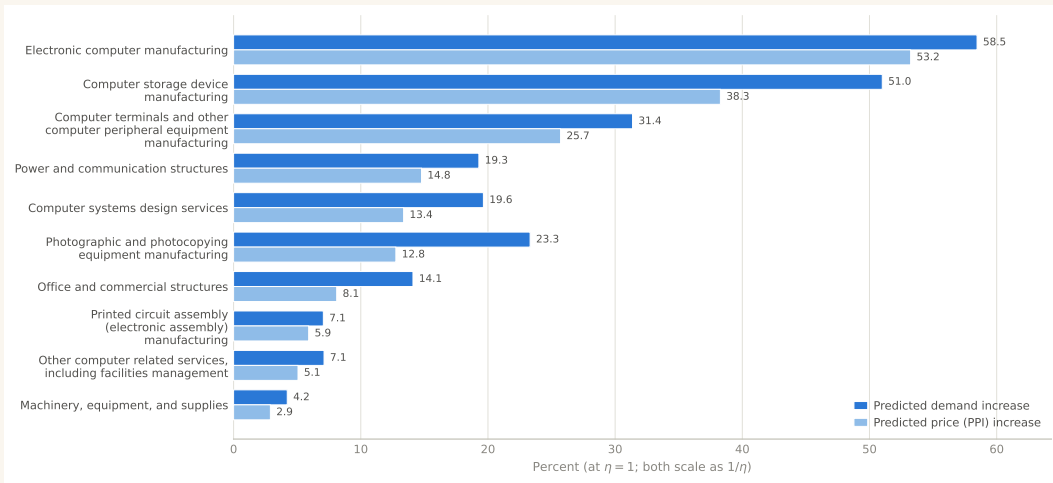
Input our demand exposure

$$\mathbf{p} = \underbrace{(\mathbf{I} - \mathbf{\Omega})^{-1} \text{diag}(\alpha_i/\eta_i)}_{\text{prices from output}} \underbrace{(\mathbf{I} - \mathbf{\Phi})^{-1}}_{\text{output from shock}} (\boldsymbol{\varepsilon})$$

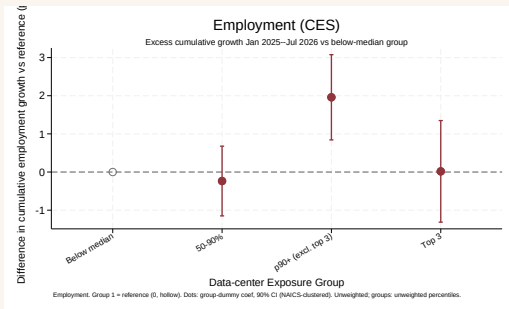
Calibration

- We use the 2017 detailed BEA input and output tables
- We assume $\eta = 1$ for value added corresponds to a total supply elasticity of around 2 similar to Boehm and Pandalai-Nayar lower estimates

Most Exposed Industries

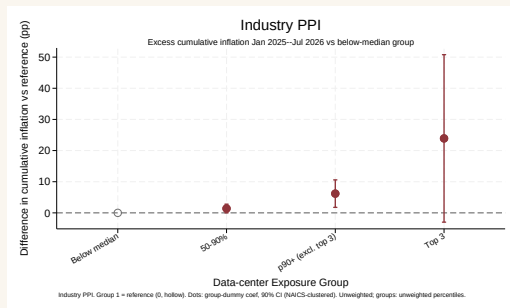


Demand exposure and employment



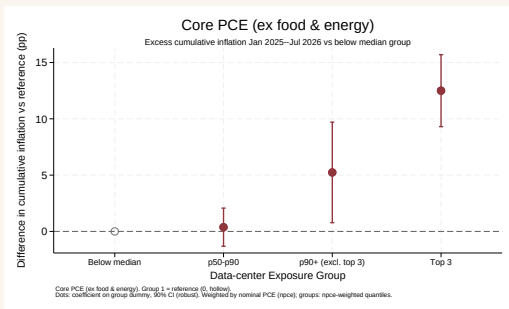
	Cumulative Employment Growth Since 2025m1	
Datcenter demand exposure (pp)	0.153 (0.094)	0.041 (0.032)
Oil exp. × oil-price growth	X	X
AI task exposure (GPT-4 beta)	X	X
Baseline log-employment (2024m12)	X	X
Pre-COVID emp trend (2017–19)	X	X
2024 emp growth	X	X
Cumulative tariff change	X	X
Weight	2024 headcount	None (equal per industry)
Observations	443	443

Demand exposure and producer prices



	Producer Price Growth Since 2025m1	
Datcenter exposure (pp)	0.749* (0.445)	0.666 (0.459)
Pre-COVID trend (2017–19)	X	X
2024 inflation	X	X
Oil exp. $\times$ oil-price growth	X	X
Cumulative tariff change	X	X
Price-change frequency (sticky price)	X	X
Weight	Net output (2017)	None
Observations	447	447

Demand exposure and consumer prices



	Price Growth Since 2025m1	
Datcenter exposure (pp)	0.470*** (0.101)	0.440*** (0.093)
Pre-COVID trend (2015–19)	X	X
2024 inflation	X	X
Oil exp. × oil-price growth	X	X
Cumulative tariff effect	X	X
Price-change frequency (sticky price)	X	X
Weight	Nominal PCE (npce)	None (equal per category)
Observations	176	176

Thank you!